AUTOMATED ATTENDANCE SYSTEM IN RURAL AREAS

Design Thinking for Innovation & Entrepreneurship (DTIE)

Section 1 | Project Batch 7

Faculty Mentor: Dr. G Sudhakar

Team Members

J. Gayathri (2620030290)

Aditi (2620030464)

A. Kusuma Sree (2620030511)

G. Pragnitha (2620040018)

B. Spoorthi (2620090043)

K. Saanvi (2620090044)

1. Vision and Mission

Vision

Every child in every village counted, present and learning – with attendance data that is accurate, instant and trusted.

Mission

Replace paper registers with a low-cost, offline-first attendance tool.

Give back teaching time lost to roll calls and paperwork.

Alert parents and administrators early when a child stops coming.

2. Problem Statement

Rural schools still track attendance on paper. Records are slow to compile, easy to fake, and rarely reach decision-makers in time to act.

Manual roll calls eat into limited teaching time – often one teacher handles several grades together.

Proxy attendance and register errors make data unreliable for schemes and audits.

Weak internet, power cuts and low-end devices break most “smart” systems built for cities.

3. Research, Facts, Market and Industry Study

Attendance in rural government primary schools (ASER)

Student attendance: 72.4% in 2018 and 75.9% in 2024.

Teacher attendance: 85.1% in 2018 and 87.5% in 2024.

Approximately 1 in 4 enrolled students is still absent on a typical day.

52.1% of government primary schools have fewer than 60 students (2024).

2/3 of Standard I–II classrooms are multigrade.

Around 90% of 14–16-year-olds report a smartphone at home – the device already exists in the village.

Monitoring works – Udaipur camera study (Seva Mandir)

Time-stamped photo attendance raised teacher attendance from 58% to 79%, adding 34 days of instruction per student per year and lifting test scores by 0.17 standard deviations.

Sources shown in the presentation: ASER 2024 via PIB (Feb 2025); IPA / J-PAL study summary.

4. Target User – Customer

Primary user: Teachers

Mark a whole class in under a minute, even with no network, on a phone they already carry.

Head teachers

Auto-generated daily and monthly registers; no manual compiling for reports.

Paying customer: Education departments

Block / district officers get live, verified attendance across hundreds of schools.

Parents and NGOs

SMS alerts on repeated absence; NGOs and CSR partners track program impact.

5. Competitors and Benchmarks

Approach	Works offline	Low cost	Stops proxy	Real-time reports	Hardware needed
Paper register	Yes	Yes	No	No	None
Fingerprint biometric device	Partly	No	Yes	Needs network	Dedicated scanner
RFID card system	Partly	Medium	Cards can be swapped	Needs network	Reader + cards
Commercial face-attendance apps	Some	Subscription	Yes	Yes	Smartphone
Time-stamped photo monitoring	Yes	Medium	Yes	Delayed	Camera
Our system	Yes	Yes	Yes	Auto-sync	Existing phone

6. Risks and Gap

Risks

Technical: QR scanning may fail if the code is damaged, blurry, or poorly printed; offline records must sync when connectivity returns.

Operational: teacher resistance, need for training, and low digital literacy.

Privacy: storing children's face / biometric data safely.

Accuracy: poor lighting and crowded classrooms affect recognition.

Gap we fill

Existing tools assume city-grade connectivity and dedicated hardware.

Biometric scanners are costly and break with no local repair support.

No single tool combines offline capture, auto-sync and a local-language UI.

Mitigation: on-device processing, encrypted storage, QR fallback, and teacher training.

7. Concept / Idea / Prototype / Product / Solution

QR ID Card Attendance – works without internet

The teacher scans each student's QR ID card using a phone to mark attendance.

QR ID scan: each student's ID card has a unique QR code.

Offline-first: attendance is saved locally and syncs when internet returns.

Dashboard for head teachers and block officers: monthly reports in one click.

SMS alerts to parents after repeated absences.

Local-language, icon-led UI for teachers with low digital literacy.

8. Images / Storyboard / Architecture

Step 1 – Scan QR ID

Student shows their QR ID card and scans it using the app.

Step 2 – Recognise

App reads the QR code and verifies the student's ID.

Step 3 – Store offline

Attendance is saved locally on the device (works without internet).

Step 4 – Auto-sync

When network is available, records are uploaded to the cloud server.

Step 5 – View and report

Dashboard shows attendance and sends SMS alerts to parents.

Technology stack (prototype)

Android app with lightweight QR code scanning.

SQLite local database + background sync.

Cloud database and web dashboard.

SMS gateway for parent alerts.

9. Lean Canvas Model

Problem

Paper registers waste teaching time; proxy attendance; no real-time data; urban tools fail offline.

Solution

Offline face-recognition app with QR fallback, auto-sync and dashboards.

Unique Value Proposition

Attendance for a whole class in under a minute – on an existing phone, with no internet.

Unfair Advantage

Built and field-tested for low-connectivity villages; local-language UI.

Customer Segments

Government and private rural schools, district education offices, NGOs, CSR programs.

Key Metrics

Schools onboarded, daily active teachers, time saved per class, absence alerts sent.

Channels

State education departments, NGO partners, teacher training camps.

Cost Structure

App development, cloud hosting, SMS charges, training and field support.

Revenue Streams

Per-school annual licence for departments, CSR / NGO-sponsored rollouts, free basic tier for single schools.

10. Capital (Financial)

Modest, student-team budget – built around our own phones, free-tier cloud tools and a 3-village pilot.

Cost head	Type	Amount (■)
App & on-device face model	One-time	80,000

Cost head	Type	Amount (■)
Pilot in 3 villages (2 weeks)	One-time	20,000
Cloud server & database	Annual	18,000
SMS gateway (~15k SMS @ ■0.20)	Annual	3,000
Field travel & teacher training	Annual	24,000
Subtotal – one-time		1,00,000
Subtotal – annual recurring		45,000
Year-1 total		1,45,000

Funding sources (Year 1)

DTIE seed grant: ■60,000.

Department incubation fund: ■45,000.

Crowdfunding / alumni CSR: ■40,000.

Total raised: ■1,45,000.

Runs on free / student-tier cloud tools. At a 10-school pilot, the recurring spend works out to about ■150 per teacher per year.

Benchmark: approximately 11¢ per extra day of school per child – Udaipur camera-study benchmark. Their study spent ■5,379 per teacher per year on film and batteries; the team-built app replaces that with its own time.

11. Roadmap / Timeline

Phase 1 – Month 1–2: Research and Design

Field interviews with teachers, empathy maps, UI wireframes.

Phase 2 – Month 3–4: Prototype

Offline app with face + QR capture, local storage and sync.

Phase 3 – Month 5–6: Pilot

Test in nearby village schools; measure time saved and accuracy.

Phase 4 – Month 7+: Scale

Partner with district offices and NGOs; add languages and analytics.

12. Conclusion

Attendance in rural schools is improving, but paper records keep data slow and unreliable.

Objective monitoring is proven to raise attendance and learning.

Our offline, phone-based system brings that monitoring to every village school at low cost.

THANK YOU

SOURCE CODE

Automated QR Attendance System (HTML, CSS and JavaScript)

```
<!doctype html>
<html lang="en">
<head>
<meta charset="utf-8">
<meta name="viewport" content="width=device-width,initial-scale=1">
<title>QR Attendance</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
<link href="https://fonts.googleapis.com/css2?family=Bricolage+Grotesque:wght@600;700&family=IBM+Plex+Mono:
wght@500&family=IBM+Plex+Sans:wght@400;500;600&display=swap" rel="stylesheet">
<script src="https://cdnjs.cloudflare.com/ajax/libs/qrcodejs/1.0.0/qrcode.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/jsQR/1.4.0/jsQR.min.js"></script>
<style>
/* Layout: dark header band with live date, then a two-column workspace.
   Left = scanner + student QR. Right = counts + roster. Stacks on phones. */
:root{
  --bg:#eef2f7; --surface:#ffffff; --ink:#0f1b2d; --muted:#5a6b80; --line:#d8e0ea;
  --accent:#2446d6; --accent-ink:#ffffff; --band:#0f1b2d; --band-ink:#eaf0fa;
  --ok:#0e7a4b; --ok-bg:#dcf4e8; --bad:#b4281f; --bad-bg:#fde4e1; --warn:#8a5a00; --warn-bg:#fdf0cf;
  --f-display:"Bricolage Grotesque","Segoe UI",system-ui,sans-serif;
  --f-body:"IBM Plex Sans","Segoe UI",system-ui,sans-serif;
  --f-mono:"IBM Plex Mono",ui-monospace,Consolas,monospace;
}
@media (prefers-color-scheme:dark){
  :root:not([data-theme="light"]){
    --bg:#0a111d; --surface:#121c2c; --ink:#e8eef8; --muted:#94a4ba; --line:#24334a;
    --accent:#7d98ff; --accent-ink:#0a111d; --band:#070c15; --band-ink:#eaf0fa;
    --ok:#52d69a; --ok-bg:#12372a; --bad:#ff8f86; --bad-bg:#3d1a18; --warn:#f2c35c; --warn-bg:#3a2d0c;
    color-scheme:dark;
  }
}
:root[data-theme="dark"]{
  --bg:#0a111d; --surface:#121c2c; --ink:#e8eef8; --muted:#94a4ba; --line:#24334a;
  --accent:#7d98ff; --accent-ink:#0a111d; --band:#070c15; --band-ink:#eaf0fa;
  --ok:#52d69a; --ok-bg:#12372a; --bad:#ff8f86; --bad-bg:#3d1a18; --warn:#f2c35c; --warn-bg:#3a2d0c;
  color-scheme:dark;
}
*{box-sizing:border-box}
html{color-scheme:light}
body{margin:0;background:var(--bg);color:var(--ink);font:15px/1.5 var(--f-body)}
button,input,select{font:inherit;color:inherit}
button{cursor:pointer}
:focus-visible{outline:3px solid var(--accent);outline-offset:2px}
.wrap{max-width:1240px;margin:0 auto;padding-left:max(16px,env(safe-area-inset-left,0px));padding-right:max(16px,env(
safe-area-inset-right,0px))}

header.band{background:var(--band);color:var(--band-ink);padding-block:18px}
header .row{display:flex;flex-wrap:wrap;gap:12px 24px;align-items:center;justify-content:space-between}
h1{font:700 1.5rem/1.1 var(--f-display);margin:0;letter-spacing:-.01em;text-wrap:balance}
header p{margin:2px 0 0;opacity:.75;font-size:.9rem}
.date{font:500 .85rem var(--f-mono);padding:6px 10px;border:1px solid rgba(255,255,255,.25);border-radius:6px;white-
space:nowrap}

main{padding-block:20px 40px;display:grid;grid-template-columns:minmax(0,380px) minmax(0,1fr);gap:20px;align-items:
start}
@media (max-width:900px){main{grid-template-columns:minmax(0,1fr)}}
.col{display:flex;flex-direction:column;gap:20px;min-width:0}

.panel{background:var(--surface);border:1px solid var(--line);border-radius:10px;padding:16px;min-width:0}
.panel h2{font:600 1.05rem var(--f-display);margin:0 0 12px;display:flex;justify-content:space-between;align-items:
center;gap:8px}
.label{font-size:.72rem;text-transform:uppercase;letter-spacing:.08em;color:var(--muted);font-weight:600}

.btn{border:1px solid var(--line);background:var(--surface);border-radius:7px;padding:8px 12px;font-weight:500}
.btn:hover{border-color:var(--accent)}
.btn.primary{background:var(--accent);border-color:var(--accent);color:var(--accent-ink)}
.btn.small{padding:4px 9px;font-size:.82rem}
.btn.danger{color:var(--bad)}
.btn[disabled]{opacity:.5;cursor:not-allowed}
.btns{display:flex;flex-wrap:wrap;gap:8px}
/* scanner */
.viewfinder{position:relative;aspect-ratio:4/3;max-width:100%;background:#050a12;border-radius:8px;overflow:hidden;
display:grid;place-items:center}
.viewfinder video{width:100%;height:100%;object-fit:cover;display:block}
.viewfinder .idle{position:absolute;inset:0;display:grid;place-items:center;padding:16px;text-align:center;color:
#9fb0c6;font-size:.9rem}
.frame{position:absolute;inset:18%;pointer-events:none;border-radius:8px;
  background:
    linear-gradient(#fff,#fff) top left/26px 3px no-repeat,linear-gradient(#fff,#fff) top left/3px 26px no-repeat,
    linear-gradient(#fff,#fff) top right/26px 3px no-repeat,linear-gradient(#fff,#fff) top right/3px 26px no-repeat,
    linear-gradient(#fff,#fff) bottom left/26px 3px no-repeat,linear-gradient(#fff,#fff) bottom left/3px 26px no-repeat,
    linear-gradient(#fff,#fff) bottom right/26px 3px no-repeat,linear-gradient(#fff,#fff) bottom right/3px 26px no-
repeat}
.result{margin-top:12px;padding:10px 12px;border-radius:8px;font-size:.92rem;border:1px solid var(--line);background:
var(--bg);overflow-wrap:anywhere}
.result.ok{background:var(--ok-bg);color:var(--ok);border-color:transparent}
.result.dup{background:var(--warn-bg);color:var(--warn);border-color:transparent}
.result.unk{background:var(--warn-bg);color:var(--warn);border-color:transparent}
.result strong{display:block;font-size:1rem}
.log{list-style:none;margin:10px 0 0;padding:0;display:flex;flex-direction:column;gap:4px;font-size:.82rem;color:var(--
muted)}
.log li{display:flex;gap:8px;justify-content:space-between;border-top:1px solid var(--line);padding-top:4px;min-width:0}
.log li span:first-child{overflow:hidden;text-overflow:ellipsis;white-space:nowrap;min-width:0}

/* qr */
.qrbox{display:grid;place-items:center;background:#fff;border-radius:8px;padding:16px;margin:12px 0}
.qrbox img,.qrbox canvas{max-width:100%;height:auto}
.qrcap{text-align:center;color:var(--muted);font-size:.85rem;margin:0 0 12px}
.qrcap b{color:var(--ink);font-size:1rem;display:block}
select,input[type=text],input[type=search]{width:100%;background:var(--surface);border:1px solid var(--line);border-
```

```
radius:7px;padding:8px 10px}
.field{display:flex;flex-direction:column;gap:4px;min-width:0}

/* stats */
.stats{display:grid;grid-template-columns:repeat(4,minmax(0,1fr));gap:12px}
@media (max-width:640px){.stats{grid-template-columns:repeat(2,minmax(0,1fr))}}
.stat{background:var(--surface);border:1px solid var(--line);border-radius:10px;padding:12px 14px}
.stat .n{font:700 2rem/1.1 var(--f-display);font-variant-numeric:tabular-nums}
.stat.present .n{color:var(--ok)} .stat.absent .n{color:var(--bad)}
.bar{height:8px;background:var(--line);border-radius:99px;overflow:hidden;margin-top:12px}
.bar i{display:block;height:100%;background:var(--ok);width:0;transition:width .3s}
@media (prefers-reduced-motion:reduce){.bar i{transition:none}}

.filters{display:grid;grid-template-columns:repeat(3,minmax(0,1fr));gap:10px;margin-bottom:12px}
@media (max-width:640px){.filters{grid-template-columns:minmax(0,1fr) minmax(0,1fr)}.filters .field:last-child{grid-
column:1/-1}}

.tablewrap{overflow-x:auto;border:1px solid var(--line);border-radius:8px}
table{border-collapse:collapse;width:100%;min-width:640px}
th,td{text-align:left;padding:9px 12px;border-bottom:1px solid var(--line);white-space:nowrap}
th{font-size:.72rem;text-transform:uppercase;letter-spacing:.08em;color:var(--muted);background:var(--bg)}
tr:last-child td{border-bottom:0}
td.roll{font-family:var(--f-mono);font-variant-numeric:tabular-nums}
td.act{display:flex;gap:6px}
.pill{display:inline-block;padding:2px 10px;border-radius:99px;font-size:.8rem;font-weight:600}
.pill.p{background:var(--ok-bg);color:var(--ok)}
.pill.a{background:var(--bad-bg);color:var(--bad)}
.empty{padding:28px 12px;text-align:center;color:var(--muted)}

.addgrid{display:grid;grid-template-columns:repeat(4,minmax(0,1fr)) auto;gap:10px;align-items:end}
@media (max-width:820px){.addgrid{grid-template-columns:repeat(2,minmax(0,1fr))}.addgrid .btn{grid-column:1/-1}}
.msg{margin:8px 0 0;font-size:.85rem;color:var(--bad);min-height:1.2em}
.tools{display:flex;flex-wrap:wrap;gap:8px;margin-top:12px}
.note{font-size:.82rem;color:var(--muted);margin:8px 0 0}
</style>
</head>
<body>

<header class="band">
  <div class="wrap row">
    <div>
      <h1>Automated QR Attendance System</h1>
      <p>Scan a student's QR code to mark them present. Everyone not scanned stays absent.</p>
    </div>
    <div class="date" id="dateLabel"></div>
  </div>
</header>

<div class="wrap">
<main>

  <!-- LEFT: scanner + QR -->
  <div class="col">
    <section class="panel" aria-labelledby="scanH">
      <h2 id="scanH">Scan QR code</h2>
      <div class="viewfinder">
        <video id="video" playsinline muted hidden></video>
        <div class="idle" id="idle">Camera is off. Press Start camera, or upload a photo of a QR code.</div>
        <div class="frame" id="frame" hidden></div>
      </div>
      <div class="btns" style="margin-top:12px">
        <button class="btn primary" id="startBtn" type="button">Start camera</button>
        <button class="btn" id="stopBtn" type="button" disabled>Stop</button>
        <label class="btn" for="imgIn" style="cursor:pointer">Scan from image</label>
        <input id="imgIn" type="file" accept="image/*" hidden>
      </div>
      <div class="result" id="result" role="status" aria-live="polite">Waiting for a scan.</div>
      <ul class="log" id="log"></ul>
      <p class="note">Reads any QR code. Student QR codes made here mark attendance. A roll number typed into a QR also
works.</p>
    </section>

    <section class="panel" aria-labelledby="qrH">
      <h2 id="qrH">Student QR code</h2>
      <div class="field">
        <label class="label" for="qrStudent">Student</label>
        <select id="qrStudent"></select>
      </div>
      <div class="qrbox" id="qrbox"></div>
      <p class="qrcap" id="qrcap"></p>
      <div class="btns">
        <button class="btn" id="dlQr" type="button">Download QR (PNG)</button>
      </div>
    </section>
  </div>

  <!-- RIGHT: counts + roster -->
  <div class="col">
    <section aria-label="Attendance counts">
      <div class="stats">
        <div class="stat"><div class="label">Total students</div><div class="n" id="sTotal">0</div></div>
        <div class="stat present"><div class="label">Present</div><div class="n" id="sPresent">0</div></div>
        <div class="stat absent"><div class="label">Absent</div><div class="n" id="sAbsent">0</div></div>
        <div class="stat"><div class="label">Attendance</div><div class="n" id="sPct">0%</div></div>
      </div>
      <div class="bar" aria-hidden="true"><i id="barFill"></i></div>
    </section>

    <section class="panel" aria-labelledby="rosterH">
      <h2 id="rosterH">Attendance list <span class="label" id="viewLabel"></span></h2>
      <div class="filters">
        <div class="field"><label class="label" for="fClass">Class</label><select id="fClass"></select></div>
        <div class="field"><label class="label" for="fSec">Section</label><select id="fSec"></select></div>
        <div class="field"><label class="label" for="fQ">Search name or roll no</label><input type="search" id="fQ"
autocomplete="off"></div>
      </div>
      <div class="tablewrap">
        <table>
```

```
          <thead><tr><th>Roll no</th><th>Student name</th><th>Class</th><th>Section</th><th>Status</th><th>Time</
th><th>Actions</th></tr></thead>
          <tbody id="tbody"></tbody>
        </table>
      </div>
      <div class="tools">
        <button class="btn" id="csvBtn" type="button">Export today (CSV)</button>
        <button class="btn danger" id="resetBtn" type="button">Reset today's attendance</button>
        <button class="btn danger" id="clearBtn" type="button">Remove all students</button>
      </div>
      <p class="note" id="sampleNote"></p>
    </section>

    <section class="panel" aria-labelledby="addH">
      <h2 id="addH">Add a student</h2>
      <form id="addForm" class="addgrid" novalidate>
        <div class="field"><label class="label" for="aName">Name</label><input type="text" id="aName"
autocomplete="off"></div>
        <div class="field"><label class="label" for="aRoll">Roll no</label><input type="text" id="aRoll"
autocomplete="off"></div>
        <div class="field"><label class="label" for="aClass">Class</label><input type="text" id="aClass" placeholder="e.
g. 10" autocomplete="off"></div>
        <div class="field"><label class="label" for="aSec">Section</label><input type="text" id="aSec" placeholder="e.g.
 A" autocomplete="off"></div>
        <button class="btn primary" type="submit">Add student</button>
      </form>
      <p class="msg" id="addMsg" role="alert"></p>
    </section>
  </div>
</main>
</div>

<script>
(function(){
"use strict";

/* ---------- helpers ---------- */
var $ = function(s){return document.querySelector(s)};
function esc(s){return String(s).replace(/[&<>"']/g,function(c){return {"&":"&amp;","<":"&lt;",">":"&gt;",'"':"&quot;",
"'":"&#39;"}[c]})}
function ls(k,def){try{var v=localStorage.getItem(k);return v?JSON.parse(v):def}catch(e){return def}}
function save(k,v){try{localStorage.setItem(k,JSON.stringify(v))}catch(e){}}
function pad(n){return String(n).padStart(2,"0")}
function todayKey(){var d=new Date();return d.getFullYear()+"-"+pad(d.getMonth()+1)+"-"+pad(d.getDate())}
function nowTime(){var d=new Date();return pad(d.getHours())+":"+pad(d.getMinutes())+":"+pad(d.getSeconds())}
function mkId(cls,sec,roll){return (cls+"-"+sec+"-"+roll).toLowerCase().replace(/\s+/g,"")}

/* ---------- data ---------- */
var K_STU="qra_students_v1", K_ATT="qra_attendance_v1", K_SAMPLE="qra_sample_v1";
var SAMPLE=[
 ["01","Aarav Reddy","10","A"],["02","Ananya Sharma","10","A"],["03","Karthik Rao","10","A"],["04","Meera Iyer","10",
"A"],["05","Rohan Verma","10","A"],
 ["01","Sneha Patel","10","B"],["02","Vikram Singh","10","B"],["03","Divya Nair","10","B"],["04","Imran Khan","10","B"],
 ["01","Lakshmi Prasad","9","A"],["02","Arjun Mehta","9","A"],["03","Pooja Kulkarni","9","A"]
].map(function(r){return {id:mkId(r[2],r[3],r[0]),roll:r[0],name:r[1],cls:r[2],sec:r[3]}});

var students=ls(K_STU,null), usingSample=ls(K_SAMPLE,null);
if(!students){students=SAMPLE.slice();usingSample=true;save(K_STU,students);save(K_SAMPLE,true)}
var att=ls(K_ATT,{});
var today=todayKey();
if(!att[today])att[today]={};

var fClass="All", fSec="All", fQ="";
var armed={};

function persist(){save(K_STU,students);save(K_ATT,att);save(K_SAMPLE,!!usingSample)}

/* ---------- filters / views ---------- */
function uniq(arr){return arr.filter(function(v,i){return arr.indexOf(v)===i})}
function sortNat(a,b){return String(a).localeCompare(String(b),undefined,{numeric:true})}
function visible(){
  var q=fQ.trim().toLowerCase();
  return students.filter(function(s){
    if(fClass!=="All"&&s.cls!==fClass)return false;
    if(fSec!=="All"&&s.sec!==fSec)return false;
    if(q&&s.name.toLowerCase().indexOf(q)<0&&s.roll.toLowerCase().indexOf(q)<0)return false;
    return true;
  }).sort(function(a,b){return sortNat(a.cls,b.cls)||sortNat(a.sec,b.sec)||sortNat(a.roll,b.roll)});
}
function renderFilters(){
  var cls=uniq(students.map(function(s){return s.cls})).sort(sortNat);
  var secs=uniq(students.map(function(s){return s.sec})).sort(sortNat);
  if(fClass!=="All"&&cls.indexOf(fClass)<0)fClass="All";
  if(fSec!=="All"&&secs.indexOf(fSec)<0)fSec="All";
  $("#fClass").innerHTML='<option value="All">All classes</option>'+cls.map(function(c){return '<option value="'+esc(
c)+'">Class '+esc(c)+'</option>'}).join("");
  $("#fSec").innerHTML='<option value="All">All sections</option>'+secs.map(function(c){return '<option value="'+esc(
c)+'">Section '+esc(c)+'</option>'}).join("");
  $("#fClass").value=fClass;$("#fSec").value=fSec;
}

/* ---------- render ---------- */
function render(){
  var list=visible();
  var present=list.filter(function(s){return att[today][s.id]}).length;
  var total=list.length, absent=total-present;
  var pct=total?Math.round(present*100/total):0;
  $("#sTotal").textContent=total;
  $("#sPresent").textContent=present;
  $("#sAbsent").textContent=absent;
  $("#sPct").textContent=pct+"%";
  $("#barFill").style.width=pct+"%";
  $("#viewLabel").textContent=(fClass==="All"?"All classes":"Class "+fClass)+" · "+(fSec==="All"?"All sections":
"Section "+fSec);

  var rows=list.map(function(s){
    var t=att[today][s.id];
    return '<tr>'+
```

```
      '<td class="roll">'+esc(s.roll)+'</td>'+
      '<td>'+esc(s.name)+'</td>'+
      '<td>'+esc(s.cls)+'</td>'+
      '<td>'+esc(s.sec)+'</td>'+
      '<td><span class="pill '+(t?'p':'a')+'">'+(t?'Present':'Absent')+'</span></td>'+
      '<td class="roll">'+(t?esc(t):'—')+'</td>'+
      '<td class="act">'+
        '<button class="btn small" data-act="qr" data-id="'+esc(s.id)+'" type="button">QR</button>'+
        '<button class="btn small" data-act="toggle" data-id="'+esc(s.id)+'" type="button">'+(t?'Mark absent':'Mark
present')+'</button>'+
        '<button class="btn small danger" data-act="del" data-id="'+esc(s.id)+'" type="button">'+(armed["del"+s.
id]?'Sure?':'Remove')+'</button>'+
      '</td></tr>';
  }).join("");
  $("#tbody").innerHTML=rows||'<tr><td colspan="7" class="empty">No students match. Add a student below or change the
filters.</td></tr>';

  $("#sampleNote").textContent=usingSample?"These are sample students. Remove them with the button above when you add
your own.":"";

  var sel=$("#qrStudent"), keep=sel.value;
  sel.innerHTML=list.map(function(s){return '<option value="'+esc(s.id)+'">'+esc(s.roll)+' · '+esc(s.name)+' ('+esc(s.
cls)+'-'+esc(s.sec)+')</option>'}).join("");
  if(keep&&list.some(function(s){return s.id===keep}))sel.value=keep;
  drawQR(sel.value);
}

/* ---------- QR generation ---------- */
function payload(s){return JSON.stringify({t:"QRATT",id:s.id,roll:s.roll,name:s.name,cls:s.cls,sec:s.sec})}
function drawQR(id){
  var box=$("#qrbox"), cap=$("#qrcap");
  box.innerHTML="";
  var s=students.filter(function(x){return x.id===id})[0];
  if(!s){cap.textContent="No student to show.";$("#dlQr").disabled=true;return}
  $("#dlQr").disabled=false;
  if(typeof QRCode==="undefined"){cap.textContent="QR library did not load. Check your internet connection and reload.";
return}
  new QRCode(box,{text:payload(s),width:200,height:200,correctLevel:QRCode.CorrectLevel.M});
  cap.innerHTML='<b>'+esc(s.name)+'</b>Roll '+esc(s.roll)+' · Class '+esc(s.cls)+' · Section '+esc(s.sec);
}
function downloadQR(){
  var id=$("#qrStudent").value;
  var s=students.filter(function(x){return x.id===id})[0];
  var src=$("#qrbox canvas");
  if(!s||!src)return;
  var c=document.createElement("canvas"), pad_=24, cap=44;
  c.width=src.width+pad_*2; c.height=src.height+pad_*2+cap;
  var x=c.getContext("2d");
  x.fillStyle="#fff";x.fillRect(0,0,c.width,c.height);
  x.drawImage(src,pad_,pad_);
  x.fillStyle="#0f1b2d";x.textAlign="center";
  x.font="600 16px sans-serif";x.fillText(s.name,c.width/2,src.height+pad_+26);
  x.font="13px sans-serif";x.fillText("Roll "+s.roll+" · Class "+s.cls+"-"+s.sec,c.width/2,src.height+pad_+44);
  var a=document.createElement("a");
  a.href=c.toDataURL("image/png");
  a.download="QR_"+s.cls+s.sec+"_"+s.roll+"_"+s.name.replace(/\s+/g,"_")+".png";
  document.body.appendChild(a);a.click();a.remove();
}

/* ---------- scanning ---------- */
var video=$("#video"), stream=null, scanning=false, scratch=document.createElement("canvas"), sctx=scratch.getContext(
"2d",{willReadFrequently:true});
var lastText="", lastAt=0, logItems=[];

function beep(ok){
  try{
    var A=window.AudioContext||window.webkitAudioContext;var a=new A();
    var o=a.createOscillator(),g=a.createGain();o.connect(g);g.connect(a.destination);
    o.frequency.value=ok?880:240;g.gain.value=.08;o.start();
    setTimeout(function(){o.stop();a.close()},ok?120:260);
  }catch(e){}
}
function setResult(kind,html){var r=$("#result");r.className="result "+kind;r.innerHTML=html}
function addLog(text,status){
  logItems.unshift({t:nowTime(),text:text,status:status});
  logItems=logItems.slice(0,6);
  $("#log").innerHTML=logItems.map(function(l){return '<li><span>'+esc(l.text)+'</span><span>'+esc(l.t)+' · '+esc(l.
status)+'</span></li>'}).join("");
}
function findStudent(text){
  var t=String(text).trim(), o=null;
  try{o=JSON.parse(t)}catch(e){}
  if(o&&o.t==="QRATT"){
    return {student:students.filter(function(s){return s.id===o.id})[0]||null,info:o};
  }
  var low=t.toLowerCase();
  var s=visible().filter(function(x){return x.roll.toLowerCase()===low})[0]
     ||students.filter(function(x){return x.id===low})[0]||null;
  return {student:s,info:null};
}
function handleScan(text){
  var now=Date.now();
  if(text===lastText&&now-lastAt<3000)return;
  lastText=text;lastAt=now;
  var f=findStudent(text), s=f.student;
  if(s){
    if(att[today][s.id]){
      setResult("dup",'<strong>'+esc(s.name)+'</strong>Already marked present at '+esc(att[today][s.id])+'. Roll '+esc(
s.roll)+', Class '+esc(s.cls)+'-'+esc(s.sec)+'.');
      addLog(s.name,"already present");beep(false);
    }else{
      att[today][s.id]=nowTime();persist();render();
      setResult("ok",'<strong>'+esc(s.name)+' marked present</strong>Roll '+esc(s.roll)+', Class '+esc(s.cls)+',
Section '+esc(s.sec)+'.');
      addLog(s.name,"present");beep(true);
    }
  }else if(f.info){
    setResult("unk",'<strong>Not on this roster</strong>'+esc(f.info.name||"Unknown")+' (roll '+esc(f.info.roll||"?")+',
```

```
 class '+esc(f.info.cls||"?")+'-'+esc(f.info.sec||"?")+'). Add this student first.');
    addLog(f.info.name||"Unknown student","not on roster");beep(false);
  }else{
    var shown=text.length>300?text.slice(0,300)+"…":text;
    setResult("unk",'<strong>QR code read, not a student</strong>Content: '+esc(shown));
    addLog(text.slice(0,40),"other QR");beep(false);
  }
}
function decode(imgData){
  if(typeof jsQR==="undefined")return null;
  var c=jsQR(imgData.data,imgData.width,imgData.height,{inversionAttempts:"attemptBoth"});
  return c&&c.data?c.data:null;
}
function tick(){
  if(!scanning)return;
  if(video.readyState===video.HAVE_ENOUGH_DATA&&video.videoWidth){
    var w=video.videoWidth,h=video.videoHeight,max=640,k=Math.min(1,max/w);
    scratch.width=Math.round(w*k);scratch.height=Math.round(h*k);
    sctx.drawImage(video,0,0,scratch.width,scratch.height);
    var text=decode(sctx.getImageData(0,0,scratch.width,scratch.height));
    if(text)handleScan(text);
  }
  requestAnimationFrame(tick);
}
function startCam(){
  if(!navigator.mediaDevices||!navigator.mediaDevices.getUserMedia){
    setResult("unk","<strong>Camera not available</strong>Open this page in Chrome, Edge, Safari or Firefox from a file
or an https link. You can still use Scan from image.");
    return;
  }
  navigator.mediaDevices.getUserMedia({video:{facingMode:{ideal:"environment"}},audio:false}).then(function(st){
    stream=st;video.srcObject=st;video.hidden=false;$("#idle").hidden=true;$("#frame").hidden=false;
    return video.play();
  }).then(function(){
    scanning=true;$("#startBtn").disabled=true;$("#stopBtn").disabled=false;
    setResult("","Camera on. Hold a QR code inside the frame.");
    tick();
  }).catch(function(e){
    stopCam();
    setResult("unk","<strong>Camera blocked</strong>Allow camera access for this page and try again, or use Scan from
image.");
  });
}
function stopCam(){
  scanning=false;
  if(stream){stream.getTracks().forEach(function(t){t.stop()});stream=null}
  video.srcObject=null;video.hidden=true;$("#idle").hidden=false;$("#frame").hidden=true;
  $("#startBtn").disabled=false;$("#stopBtn").disabled=true;
}
function scanFile(file){
  if(!file)return;
  var url=URL.createObjectURL(file), img=new Image();
  img.onload=function(){
    var k=Math.min(1,1200/Math.max(img.width,img.height));
    scratch.width=Math.round(img.width*k);scratch.height=Math.round(img.height*k);
    sctx.drawImage(img,0,0,scratch.width,scratch.height);
    URL.revokeObjectURL(url);
    var text=decode(sctx.getImageData(0,0,scratch.width,scratch.height));
    lastText="";
    if(text)handleScan(text);else setResult("unk","<strong>No QR code found</strong>Try a sharper, closer photo with
the whole code visible.");
  };
  img.onerror=function(){URL.revokeObjectURL(url);setResult("unk","<strong>Could not open that image</strong>")};
  img.src=url;
}

/* ---------- actions ---------- */
function twoStep(key,btn,label,fn){
  if(armed[key]){delete armed[key];fn();return}
  armed[key]=true;btn.textContent=label;
  setTimeout(function(){if(armed[key]){delete armed[key];render();resetBtns()}},3000);
}
function resetBtns(){$("#resetBtn").textContent="Reset today's attendance";$("#clearBtn").textContent="Remove all
students"}

$("#tbody").addEventListener("click",function(e){
  var b=e.target.closest("button[data-act]");if(!b)return;
  var id=b.getAttribute("data-id"), act=b.getAttribute("data-act");
  if(act==="toggle"){
    if(att[today][id])delete att[today][id];else att[today][id]=nowTime();
    persist();render();
  }else if(act==="qr"){
    $("#qrStudent").value=id;drawQR(id);
    $("#qrH").scrollIntoView({behavior:"smooth",block:"start"});
  }else if(act==="del"){
    if(armed["del"+id]){
      delete armed["del"+id];
      students=students.filter(function(s){return s.id!==id});
      Object.keys(att).forEach(function(d){delete att[d][id]});
      persist();renderFilters();render();
    }else{
      armed["del"+id]=true;render();
      setTimeout(function(){if(armed["del"+id]){delete armed["del"+id];render()}},3000);
    }
  }
});
$("#resetBtn").addEventListener("click",function(){
  twoStep("reset",this,"Click again to confirm",function(){att[today]={};persist();render();resetBtns();setResult("",
"Today's attendance was reset.")});
});
$("#clearBtn").addEventListener("click",function(){
  twoStep("clear",this,"Click again to confirm",function(){students=[];att={};att[today]={};usingSample=false;persist();
renderFilters();render();resetBtns()});
});
$("#csvBtn").addEventListener("click",function(){
  var list=visible();
  var lines=[["Roll no","Name","Class","Section","Status","Time"]].concat(list.map(function(s){
    var t=att[today][s.id];return [s.roll,s.name,s.cls,s.sec,t?"Present":"Absent",t||""];
```

```
  }));
  var csv=lines.map(function(r){return r.map(function(v){return '"'+String(v).replace(/"/g,'""')+'"'}).join(",")}).join(
"\r\n");
  var a=document.createElement("a");
  a.href=URL.createObjectURL(new Blob([csv],{type:"text/csv"}));
  a.download="attendance_"+today+".csv";
  document.body.appendChild(a);a.click();a.remove();
});
$("#addForm").addEventListener("submit",function(e){
  e.preventDefault();
  var name=$("#aName").value.trim(), roll=$("#aRoll").value.trim(), cls=$("#aClass").value.trim(), sec=$("#aSec").value.
trim().toUpperCase();
  var msg=$("#addMsg");
  if(!name||!roll||!cls||!sec){msg.textContent="Fill in name, roll no, class and section.";return}
  var id=mkId(cls,sec,roll);
  if(students.some(function(s){return s.id===id})){msg.textContent="Roll no "+roll+" already exists in class "+cls+"-
"+sec+".";return}
  if(usingSample){students=[];att={};att[today]={};usingSample=false}
  students.push({id:id,roll:roll,name:name,cls:cls,sec:sec});
  persist();msg.textContent="";
  $("#aName").value="";$("#aRoll").value="";
  fClass=cls;fSec=sec;fQ="";$("#fQ").value="";
  renderFilters();render();
  $("#qrStudent").value=id;drawQR(id);
});
$("#fClass").addEventListener("change",function(){fClass=this.value;render()});
$("#fSec").addEventListener("change",function(){fSec=this.value;render()});
$("#fQ").addEventListener("input",function(){fQ=this.value;render()});
$("#qrStudent").addEventListener("change",function(){drawQR(this.value)});
$("#dlQr").addEventListener("click",downloadQR);
$("#startBtn").addEventListener("click",startCam);
$("#stopBtn").addEventListener("click",stopCam);
$("#imgIn").addEventListener("change",function(){scanFile(this.files[0]);this.value=""});
window.addEventListener("pagehide",stopCam);

/* ---------- boot ---------- */
$("#dateLabel").textContent=new Date().toLocaleDateString(undefined,{weekday:"long",day:"numeric",month:"long",year:
"numeric"});
renderFilters();render();
window.__qra={findStudent:findStudent};
})();
</script>
</body>
</html>
```